

READER'S MAP — orientation and glossary for the GRUT working corpus

For the physicist reader arriving from outside. The corpus (Books I–X) is written in the program's internal provenance vocabulary; nothing in that vocabulary is decorative, but none of it should be a barrier. This map is the translation layer. Statuses: `books/CORPUS_CHARTER.md` . Front door: `BOOK_I_FOUNDATIONS.md` .

GRUT Books — Working Edition v1.0 · D. Ryan Grover · 2026-09-07 · release `zenodo-v1.0` · Markdown is the authoritative source; the PDF is generated from it. Drafted from the frozen record by the RAI builder (Claude, Anthropic) under owner direction — this edition is a publishing pass only: formatting, navigation, and metadata; no claim strengthened or weakened for publication.

The framework in six sentences

- 1. Core model.** GRUT treats the gravitational vacuum as a responsive medium: physics is organized through retarded response kernels K_R and noise kernels N in a Schwinger–Keldysh influence action, constrained by causality (Kramers–Kronig), equilibrium detailed balance (KMS/FDT), passivity, and noise positivity.
- 2. Inputs.** Every primitive is declared and priced: the medium stance itself, the system/bath split, the TT-only projector choice, the $x = 0$ point, the Born measure, the temporal orientation, the finite-memory bet (with its reversal on its face).
- 3. Derived, at scope.** A one-loop TT kernel with $H^1 = 0$ identically (and a genuinely new even-degree cancellation identity behind it), an $s = 5$ spectral law, an exact de Sitter tail $H^2/4\pi$, passivity/cone theorems, a linear-universe response no-go.
- 4. Recovered, by identity.** GR at zero-memory collapse and standard open-system physics — compatibility on declared inputs, not correspondence evidence.
- 5. Ruled out or refused, by the program's own instruments.** The $\mu = 4/3$ endpoint (self-excluded), the observable tensor-friction route (computed REFUSE, the licensed value 62.7 orders below the observational slot), the $\omega \approx 3.4H$ regime (unaskable at current declarations), and every audited resurrection of the undeclared-input pattern.
- 6. What would count as a prediction.** A pre-registered quantitative consequence, not encoded in the inputs, surviving EFT/modified-gravity degeneracy, produced through a design gate (Book IX). The PREDICTED set is currently empty, and the corpus says so.

Glossary of internal labels

term	meaning
the register	<code>provenance/claims.json</code> — 74 claims, each with tier, dependencies, sources, an overturning computation, and a priced ledger delta; the program's single source of claim-status truth

term	meaning
rung1...rung9	the historical derivational ladder's steps (in-in formalism, KMS gate, single-pole kernel, KK/Love, ..., cosmology); the ladder is frozen — the rungs survive as register nodes, not as a live derivation program
tier	a register node's native standing: shown / derived-pending / to-derive / assumed / postulate (crosswalk to display statuses in the charter)
banking / banked	admitting a result into the register after gate checks; "nothing banks" = a document that changes no register state
W-0	"computed and reported, not banked" — the standing rule for work products of this period
+N / priced	every declared input costs ledger points at its point of entry; the running net measures how much is assumed
laundering	importing an assumption without pricing it — the program's cardinal sin; "relocation reported as discharge" is its signature shape
fence	a recorded scope restriction or non-promotion rule ("a detected $\Xi_0 \neq 1$ could never confirm GRUT")
u1, u3, u4	universality/uniqueness question nodes: u1 = the influence-functional form is generic (no credit for the form); u3/u4 = the system/bath split's origin and dissolution questions (fenced open)
x, x_no_pin	x = the scalar-to-tensor kernel ratio; the x_no_pin theorem: the admissible set is an amplitude-homogeneous cone — the action pins no ratio
p_tt_ansatz	the TT-only projector restriction — interrogated and adjudicated CHOSEN (not forced)
Tier-4 / the K_R contract	the campaign that computed the one-loop retarded TT self-energy on declared inputs at flat contract scope, $\omega \gg H$
H¹ = 0	the derived vanishing of the first-order-in-H absorptive correction — a four-channel cancellation with an even-degree "ladder" mechanism (Theorem II)
ROOT-0 / ROOT-1	the kernel-origin audits; ROOT-1 established the $\omega \lesssim 3.4H$ regime is UNASKABLE at current declarations (four obstructions O1–O4)
the five-angle interrogation	the adversarial procedure that adjudicated p_tt_ansatz as CHOSEN, unanimously
RAI instruments	adversarial multi-agent audits (Resurrection, Gorilla-T1, Dialectic Chamber, Final Boss) — each a pre-registered, de-pinned instrument whose verdicts are quoted at recorded strength
de-pinned	the instrument standard: gates assert well-formedness only, never which verdict passes — the answer to the recurring "pass-label" defect
the Γ_T gate / closure	the prediction-gate exercise on gravitational-wave tensor friction: design-first protocol, computed REFUSE, licensed value 62.7 orders below the bound
the freeze / stopping rule	GRUT_PROGRAM_FREEZE.md : no new foundational generation merely because the previous one failed; reopening requires an independently motivated principle
quiet-or-faint	the banked magnitude verdict on the gravitational-decoherence hypothesis: consistent with all data by being small

term	meaning
the half-line/KMS alignment	the single surviving unexplained temporal datum: which side of a KMS weight the spectral half-line decays toward — every audited arrow-of-time closure consumed it
B-INPUT-RELOCATION	an audited verdict class: a construction moves the unexplained input somewhere else rather than discharging it
the match temptation	the recorded rule that a computed number landing near an observational bound is scrutinized hardest, never celebrated

Where to start, by reader

- "What is GRUT claiming?" → Book I, then Book IX's entry conditions.
- "Show me the mathematics." → Book II, then Book IV for the computed kernel.
- "What did it get wrong?" → Book I §3, Book X's failure-history section — presented as scientific content, not apologetics.
- "Is any of this testable?" → Book IX, including why the answer today is "not yet, and here is exactly what stands between the framework and a test."